IST 659 Final Project: Passenger Train Management System

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Project Overview

Train Management Database System

Problem Statement

- Lack of Timely Communication
- Limited Transparency Regarding Delays and Service Disruptions
- Insufficient Train and Track Maintenance
- Complicated Ticket Booking Processes and Website Downtime
- Misunderstanding of Routes Among Passengers

Project Goals

To create a system that will streamline and simplify the management of passenger trains, while enhancing transparency between administrators, operators, and passengers.

01

Intuitive Route Maps and Train Schedules

02

Real-Time Train Tracking

03

Alerts and Notifications of Delays and Disruptions

04

Reliable Ticket Booking via Normalization and Transactions

05

Database-Driven Proactive Maintenance

Data Requirements

Entities and Attributes

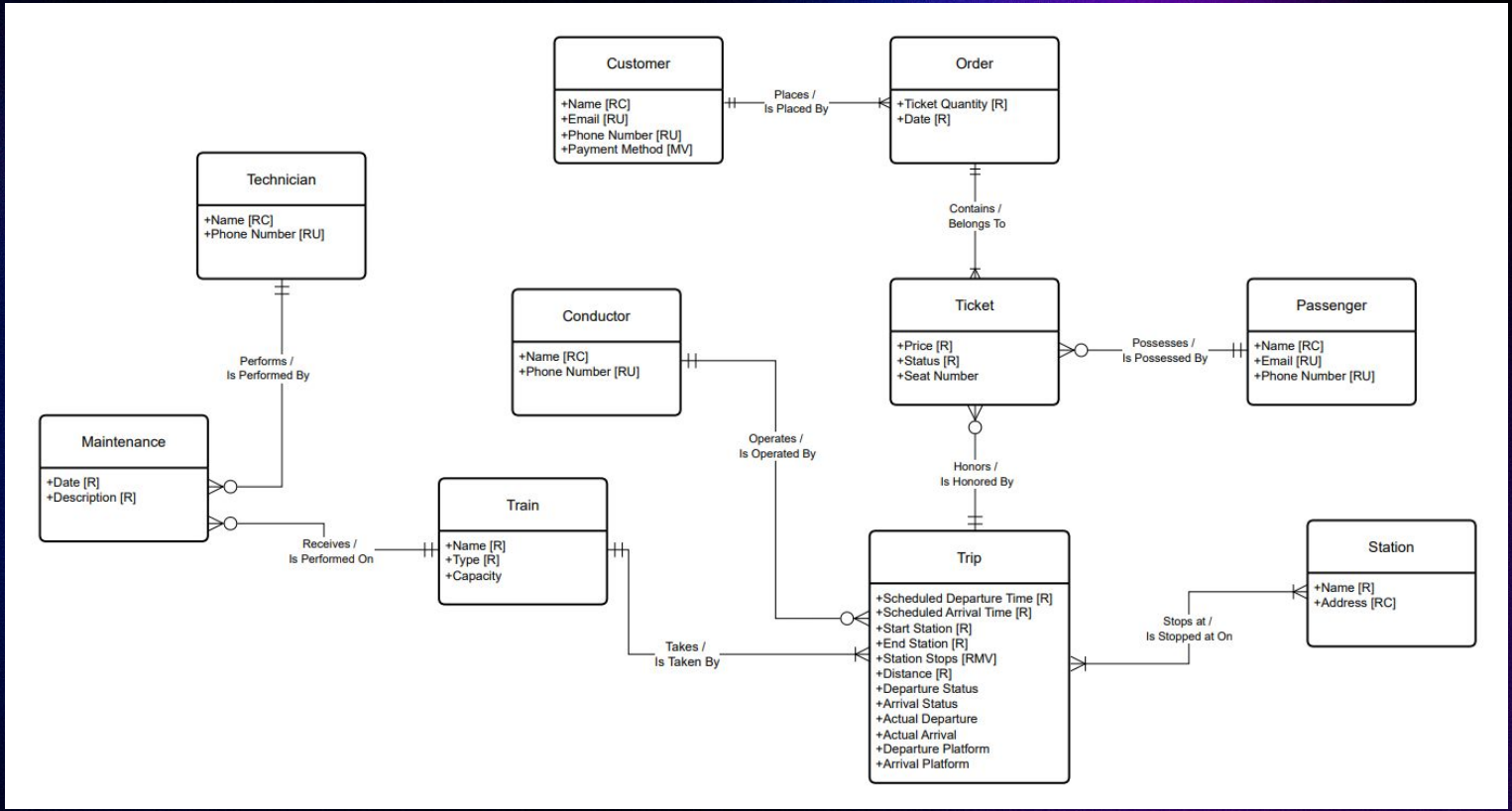
Entity	Attributes	Properties	Description
Train	Name	R	Name of the train
	Type	R	Passenger or Freight
	Capacity	R	Number of passengers the train can hold
Station	Name	R	Name of the train station
	Address	RC	Address of the train station
Trip	Scheduled Departure Time	R	Time the train is scheduled to depart from the trip's start station
	Scheduled Arrival Time	R	Time the train is scheduled to arrive at the trip's end station
	Start Station	R	Station where the train starts
	End Station	R	Station where the train stops last
	Station Stops	RMV	Stations where the train stops
	Distance	R	Distance in miles of the entire route
	Departure Status		Whether the train is departing on time, early, or late from the trip's start station
	Arrival Status		Whether the train is arriving on time, early, or late to the trip's end station
	Actual Departure		Time the train actually departs from the trip's start station
	Actual Arrival		Time the train actually arrives at the trip's end station
	Departure Platform		Platform where the train will board at the trip's start station
	Arrival Platform		Platform where the train will stop at the trip's end station
Customer	Name	RC	Name of the customer
	Email	RU	Email of the customer
	Phone Number	RU	Phone number of the customer
	Payment Method	MV	Customer's method of payment, such as credit card, debit card, or gift card
Passenger	Name	RC	Name of the passenger
	Email	RU	Email of the passenger
	Phone Number	RU	Phone number of the passenger
Ticket	Price	R	Price of the ticket
	Status	R	Whether the ticket has been purchased or is still for sale
	Seat Number		Seat number assigned to the passenger who purchases the ticket
Order	Ticket Quantity	R	Number of tickets in an order
	Date	R	Date the order is placed
Conductor	Name	RC	Name of the conductor
	Phone Number	RU	Phone number of the conductor
Maintenanc	Date	R	Date of maintenance activity
	Description	R	Brief description of the maintenance that occurred
Technician	Name	RC	Name of the Technician
	Phone Number	RU	Phone number of the Technician

Data Requirements

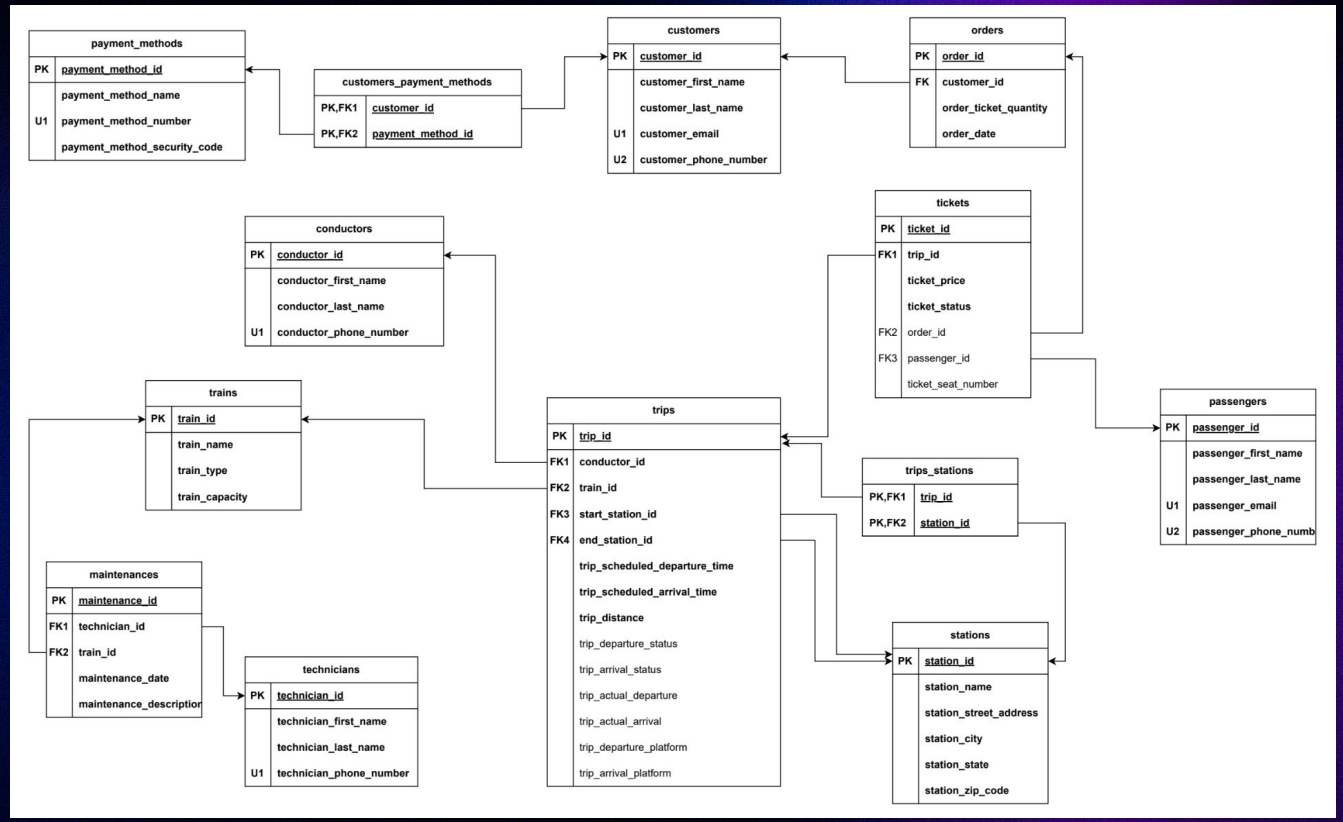
Relationships

Relationship	Entitiy	Rule	Min	Max	Entity
Train-Trip	Train	Takes	1	M	Trip
	Trip	Is taken by	1	1	Train
Train-Maintenance	Train	Receives	0	M	Maintenance
	Maintenance	Is performed on	1	1	Train
Station-Trip	Station	Is stopped at on	0	M	Trip
	Trip	Stops at	1	M	Station
Trip-Conductor	Trip	Is operated by	1	1	Conductor
	Conductor	Operates	0	M	Trip
Trip-Ticket	Trip	Honors	0	M	Ticket
	Ticket	Is valid for	1	1	Trip
Passenger-Ticket	Passenger	Possesses	1	1	Ticket
	Ticket	Is possessed by	1	1	Passenger
Customer-Order	Customer	Places	1	M	Order
	Order	Is placed by	1	1	Customer
Order-Ticket	Order	Contains	1	M	Ticket
	Ticket	Belongs to	1	1	Order
Maintenance-Technician	Maintenance	Is performed by	1	1	Technician
	Technician	Performs	0	M	Maintenance

Conceptual Model



Logical Model



User Stories

Customer

As a customer, I should be able to purchase multiple tickets so I can take a trip with other people.

Passenger

As a passenger, I should be able to view the departure and arrival status of my train, so I can know if it is delayed.

Conductor

As a conductor, I should be able to view what trips I am assigned to so I can know when I am working and where to be.

Up-Down Script

```sql
create table customers (
    customer_id int identity not null,
    customer_first_name varchar(50) not null,
    customer_last_name varchar(50) not null,
    customer_email varchar(100) not null,
    customer_phone_number varchar(10) not null,
    constraint pk_customers_customer_id primary key (customer_id),
    constraint u_customers_customer_email unique (customer_email),
    constraint u_customers_customer_phone_number unique (customer_phone_number),
    constraint ck_customers_customer_phone_digits check (customer_phone_number not like '%[^0-9]%')
)
```

```sql
create table payment_methods (
    payment_method_id int identity not NULL,
    payment_method_type varchar(20) not null,
    payment_method_number varchar(19) not NULL,
    payment_method_security_code varchar(4) not NULL,
    payment_method_expiry varchar(5) not null,
    constraint pk_payment_method_payment_method_id primary key (payment_method_id),
    constraint u_payment_method_payment_method_number unique (payment_method_number)
)
```

Up-Down Script

```sql
create table trips (
    trip_id int identity not null,
    conductor_id int not null,
    train_id int not null,
    start_station_id int not null,
    end_station_id int not null,
    trip_scheduled_departure_time datetime not null,
    trip_scheduled_arrival_time datetime not null,
    trip_distance_miles decimal(6,2) not null,
    trip_departure_status varchar(20) null,
    trip_arrival_status varchar(20) null,
    trip_actual_departure datetime null,
    trip_departure_platform int null,
    trip_arrival_platform int null,
    constraint pk_trips_trip_id primary key (trip_id),
    constraint ck_trips_estimated_arrival_departure_times check (trip_scheduled_arrival_time > trip_schedul
    constraint ck_trips_actual_arrival_departure_times check (trip_actual_departure is null or trip_actual_
    constraint ck_trips_trip_distance_miles check (trip_distance_miles > 0),
    constraint ck_trips_departure_status_options check (trip_departure_status is null or trip_departure_sta
    constraint ck_trips_arrival_status_options check (trip_arrival_status is null or trip_arrival_status in
)
```

```sql
alter table trips
add constraint fk_trips_conductor_id
    foreign key (conductor_id)
    references conductors(conductor_id)

alter table trips
add constraint fk_trips_train_id
    foreign key (train_id)
    references trains(train_id)

alter table trips
add constraint fk_trips_start_station_id
    foreign key (start_station_id)
    references stations(station_id)

alter table trips
add constraint fk_trips_end_station_id
    foreign key (end_station_id)
    references stations(station_id)
```

```sql
create table maintenances (
    maintenance_id int identity not null,
    technician_id int not NULL,
    train_id int not null,
    maintenance_date datetime not null,
    maintenance_description varchar(500) not null,
    constraint pk_maintenances_maintenance_id primary key (maintenance_id)
)
```

Up-Down Script

```sql
create table trips_stations (
    trip_id int not null,
    station_id int not null,
    constraint pk_trips_stations primary key (trip_id, station_id)
)

create table tickets (
    ticket_id int identity not null,
    trip_id int not null,
    ticket_price money not null,
    ticket_status varchar(20) not null,
    order_id int null,
    passenger_id int null,
    ticket_seat_number varchar(4) null,
    constraint pk_tickets_ticket_id primary key (ticket_id),
    constraint ck_tickets_ticket_price check (ticket_price >= 0)
)
```

Data Entry

```sql
insert into stations
    (station_name, station_street_address, station_city, station_state, station_zip_code)
VALUES
    ('Syracuse Walsh Regional Transportation Centre', '1 Walsh Cir', 'Syracuse', 'NY', '13208'),
    ('Louise M. Slaughter Rochester Station', '320 Central Ave', 'Rochester', 'NY', '14605'),
    ('Albany-Rensselaer Station', '525 East Street', 'Rensselaer', 'NY', '12144'),
    ('Buffalo Central Terminal', '495 Paderewski Drive', 'Buffalo', 'NY', '14212'),
    ('South Station', '700 Atlantic Ave', 'Boston', 'MA', '02110'),
    ('Grand Central Station', '49 East 42nd Street', 'New York City', 'NY', '10017'),
    ('Newark Penn Station', '1 Raymond Plaza West & Market Street', 'Newark', 'NJ', '07102'),
    ('Hartford Union Station', '1 Union Place', 'Hartford', 'CT', '06103'),
    ('Concord Transportation Centre', '30 Stickney Ave', 'Concord', 'NH', '03301')
```

```sql
insert into passengers
    (passenger_first_name, passenger_last_name, passenger_email, passenger_phone_number)
    VALUES
        ('Brealin', 'Redecker', 'blredeck@syr.edu', '6301648453'),
        ('Yida', 'Sun', 'ysun217@syr.edu', '1284026491'),
        ('Alexa', 'Lotano', 'alotano@syr.edu', '7102541826'),
        ('Joseph', 'Crimmer', 'jpcrimme@syr.edu', '1234567890'),
        ('Francis', 'Ewing','francisrules@gmail.com', '3364363961'),
        ('Chris', 'McCann', 'mccannchris@outlook.com', '4087929690'),
        ('Judith', 'Velazquez', 'JudithTVelazquez@armyspy.com', '8317636873')
```

Data Entry

```sql
insert into maintenances
    (technician_id, train_id, maintenance_description)
    VALUES
        (3, 1, 'Monthly inspection. Passed.'),
        (3, 2, 'Monthly inspection. Passed.'),
        (3, 3, 'Monthly inspection. Failed. Needs replacement grind rails.'),
        (2, 4, 'Monthly inspection. Passed.'),
        (1, 5, 'Monthly inspection. Failed. Needs replacement engine.'),
        (1, 5, 'Replaced engine, train should function as intended now.')
```

```sql
insert into trains
    (train_name, train_type, train_capacity)
    VALUES
        ('Zephyr', 'Passenger', '250'),
        ('Bullet', 'High-Speed Rail', '500'),
        ('Chief', 'Passenger', '300'),
        ('Thomas', 'Overnight Passenger', '300'),
        ('Percy', 'Overnight Passenger','250')
```

```sql
insert into trips
    (conductor_id, train_id, start_station_id, end_station_id, trip_scheduled_departure_time, trip_scheduled_arrival_time, trip_distance_miles)
    VALUES
        (3, 2, 4, 3, '2025-12-07 14:30:00', '2025-12-07 18:00:00', 291.32), --Buffalo to Albany
        (1, 4, 2, 9, '2025-12-11 17:00:00', '2025-12-12 02:45:00', 369.05), --Rochester to Concord
        (2, 1, 1, 5, '2025-12-12 10:45:00', '2025-12-12 20:00:00', 313.17) --Syracuse to Boston
```

Procedure

Role: Encapsulate complex business workflows into secure, callable routines.

Key Feature: Transactional Integrity (ACID)

- Used `BEGIN TRAN` and `TRY/CATCH` blocks to ensure operations are atomic ("all-or-nothing").

Implementation:

- **`p_update_trip_status`:** Updates departure/arrival times with built-in validation logic (e.g., ensuring arrival time > departure time).
- **`p_quick_purchase_ticket`:** A secure booking process that handles validation, insertion, and error rollback in a single transaction.

Function

Role: Create reusable code snippets for calculations and complex views.

Key Feature: Abstraction

- Hides complex SQL logic (like multi-table joins or date math) behind simple function calls.

Implementation:

- **Scalar Function (`f_get_scheduled_duration_minutes`):** Calculates trip duration on-the-fly, eliminating redundant date math in queries.
- **Table-Valued Function (`f_get_trip_station_names`):** Returns a clean list of all station stops for a journey, abstracting away necessary JOINs.

Trigger

Role: Automatically react to data changes to enforce business rules and maintain consistency.

Key Feature: Proactive & Reactive Safety

- Acts as a final "safety net" at the database level, regardless of application logic.

Implementation:

- **Reactive Automation (AFTER UPDATE Trigger):** If a Trip is cancelled, automatically cancel all associated tickets to ensure data consistency.
- **Proactive Enforcement (INSTEAD OF INSERT Trigger):** Prevents booking tickets for past trips by intercepting the insert and checking the date before execution.

Thank you!